

Assignment 3: Take a sample dataset and demonstrate the use of PCA technique.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
data = load_breast_cancer()
X = data.data
y = data.target

print("Original shape:", X.shape)
```

Original shape: (569, 30)

```
#Standardize Data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
#PCA Applied
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

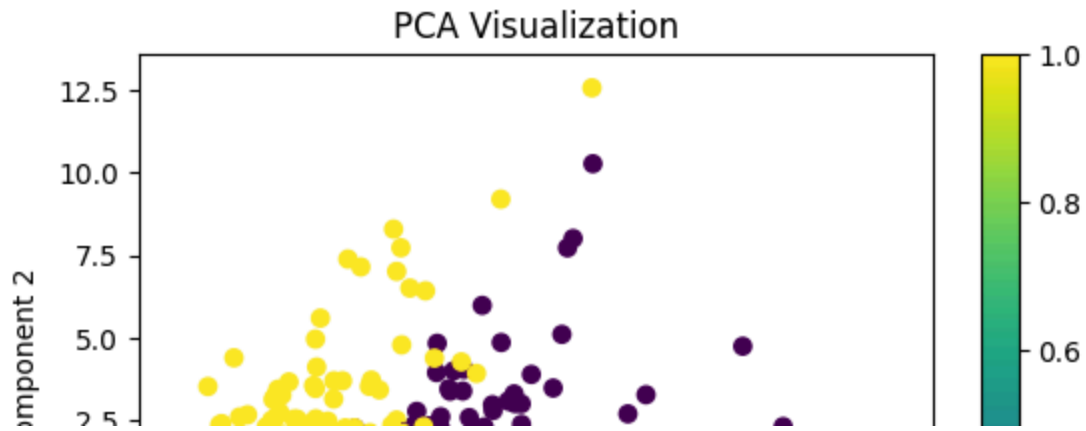
print("Reduced shape:", X_pca.shape)
```

Reduced shape: (569, 2)

```
print("Explained variance ratio:", pca.explained_variance_ratio_)
print("Total variance retained:", np.sum(pca.explained_variance_ratio_))
```

Explained variance ratio: [0.44272026 0.18971182]
Total variance retained: 0.6324320765155944

```
plt.scatter(X_pca[:,0], X_pca[:,1], c=y, cmap='viridis')
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA Visualization")
plt.colorbar()
plt.show()
```



```
#Performance Comparision
# Split original data
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)

# Logistic Regression (original)
model_original = LogisticRegression(max_iter=1000)
model_original.fit(X_train, y_train)
acc_original = accuracy_score(y_test, model_original.predict(X_test))

# PCA data
X_train_pca, X_test_pca, _, _ = train_test_split(
    X_pca, y, test_size=0.2, random_state=42
)

model_pca = LogisticRegression(max_iter=1000)
model_pca.fit(X_train_pca, y_train)
acc_pca = accuracy_score(y_test, model_pca.predict(X_test_pca))

print("Accuracy without PCA:", acc_original)
print("Accuracy with PCA:", acc_pca)
```

Accuracy without PCA: 0.9736842105263158
Accuracy with PCA: 0.9912280701754386

